

Introduction to Data Science and Engineering

- Course Overview



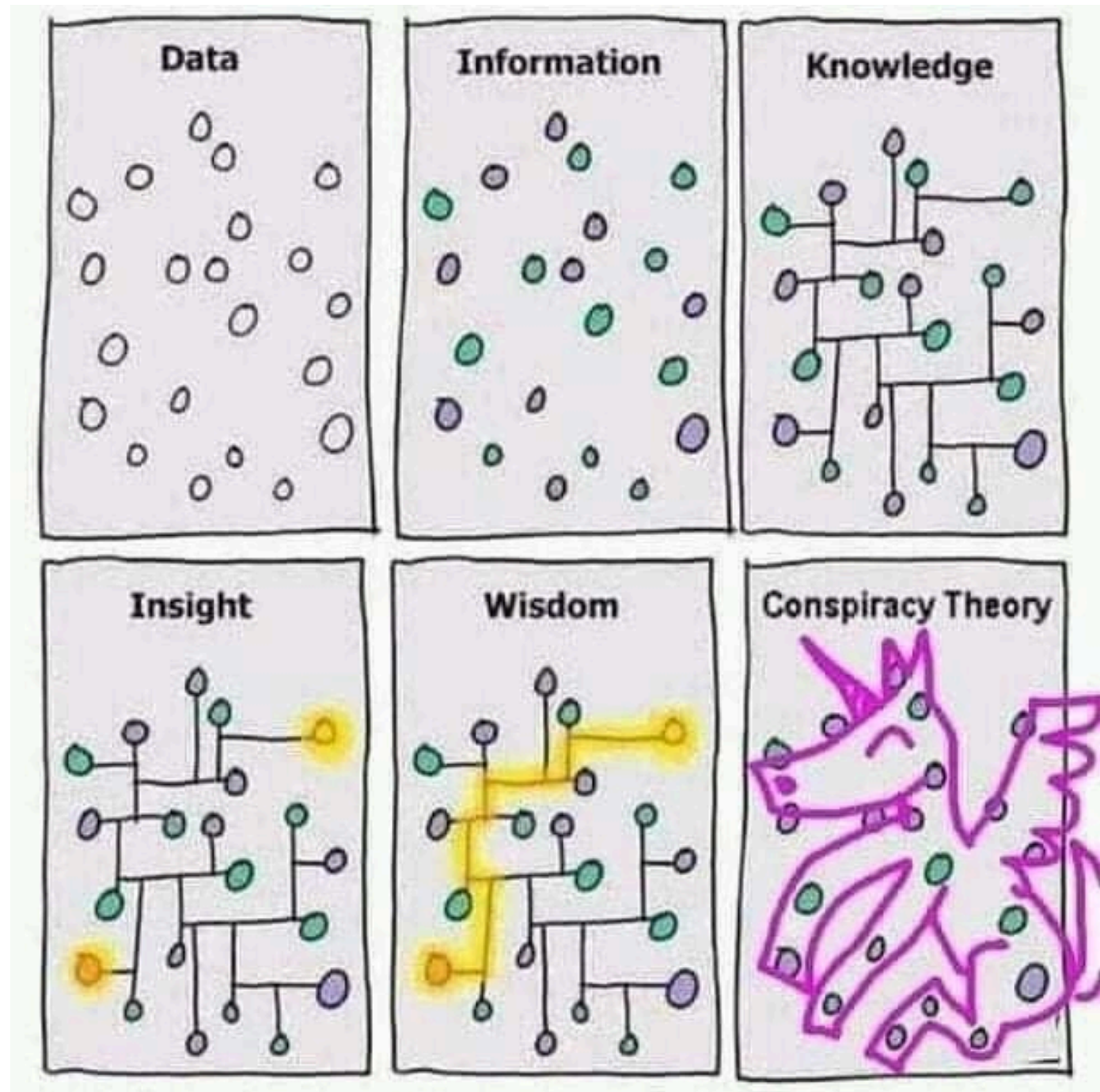
Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

RB Luo / 羅銳邦

School of Computing and Data Science
University of Hong Kong

Some slides retrofitted by Michael Wu

Things that start
with data



First thing first, what's data science?

ChatGPT 5



What is data science? Answer this question in 50 words.

Key points here:

- Extract **insights from data**
- **Interdisciplinary**

Data science is the interdisciplinary field that uses statistical methods, algorithms, and computational tools to extract meaningful insights from structured and unstructured data. It combines mathematics, programming, and domain expertise to analyze patterns, build predictive models, and support decision-making across business, science, and technology.



The term data scientist was first used by Jeff Hammerbacher and DJ Patil to describe a new job title. They searched for individuals with a new set of critical skills such as mathematics, statistics, computer science, machine learning, and business knowledge.

Is data science a real science?

Criticism: *"Big data analysis is mainly: garbage in, garbage out..."*

Criticism: *"Data science can find anything in a large amount of data. By subsetting the data and building new features, they can prove anything. As Ronald H. Coase said: "If you torture the data long enough, it will confess."*

Criticism: *"Data scientists can only build observational studies. They might spot a few correlations, but can't say anything about the underlying causes."*

...

What could make data science **more scientific**?

- Uphold **scientific values**: emphasize transparency and reproducibility
- Correct **methodology**: never presume your result, design your experiments/tests rigorously, acknowledge limitations



Ludovic Benistant

Dec 15, 2016 · 4 min read · Listen



Is Data Science A Real Science?

As I hear criticisms about how data science is unscientific, I would like to clarify what is science, and to show how data scientists can answers these common criticisms about their field.

<https://towardsdatascience.com/is-data-science-a-real-science-2920bb2529aa>

PDF of the article provided in Moodle



Yes, data scientist is a good job

50 Best Jobs in America

Best Places to Work

Top CEOs

Best Jobs

Best Cities for Jobs

Highest Paying Jobs

 [Share](#)

2016 ▾

United States ▾

Job Title	Median Base Salary	Job Satisfaction	Job Openings
#1 Data Scientist	\$116,840	4.1/5	1,736

[View Jobs](#)

But if you think data scientist is still in a stable uptrend, you made a typical data scientist mistake

50 Best Jobs in America

2022

#1	Enterprise Architect	\$144,997	4.1/5	14,021	View Jobs
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#2	Full Stack Engineer	\$101,794	4.3/5	11,252	View Jobs
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#3	Data Scientist	\$120,000	4.1/5	10,071	View Jobs
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2021

#1	Java Developer	\$90,830	4.2/5	10,103	View Jobs
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#2	Data Scientist	\$113,736	4.1/5	5,971	View Jobs
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#1	Front End Engineer	\$105,240	3.9/5	13,122	View Jobs
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2020

#2	Java Developer	\$83,589	3.9/5	16,136	View Jobs
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#3	Data Scientist	\$107,801	4.0/5	6,542	View Jobs
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2019

#1	Data Scientist	\$108,000	4.3/5	6,510	View Jobs
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2018

#1	Data Scientist	\$110,000	4.2/5	4,524	View Jobs
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2017

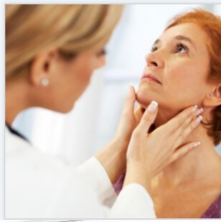
#1	Data Scientist	\$110,000	4.4/5	4,184	View Jobs
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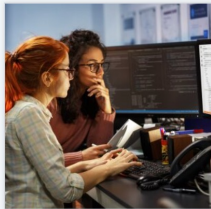
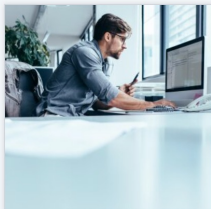
2016

#1	Data Scientist	\$116,840	4.1/5	1,736	View Jobs
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You think that's it?

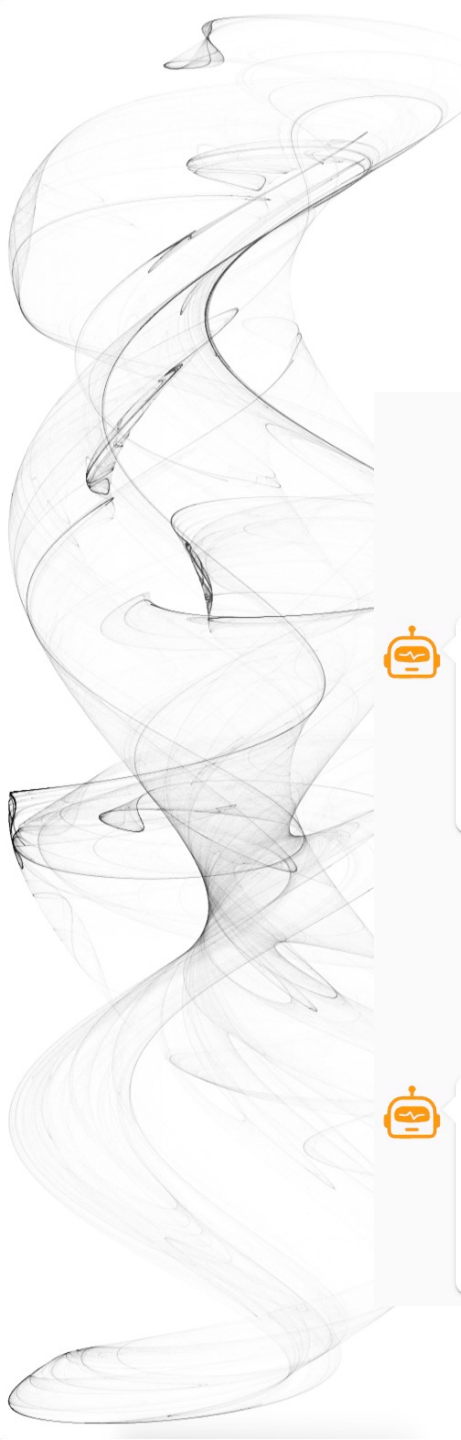


	Nurse Practitioner #1 in 100 Best Jobs Nurse practitioners perform a lot of the same duties as physicians; they perform physical exams, order lab tests and analyze results, prescribe medicines and authorize treatments. These professionals usually have a specialty, such as adult and geriatric health, pediatric health or mental health. Read More »	Projected Jobs 128,400 Median Salary \$129,210 Education Needed Master's
	Financial Manager #2 in 100 Best Jobs If financial managers were doctors, the financial success of their organization would be a testimony to their treatment. These business professionals create financial reports, coordinate investment activity and develop long-term financial strategies for companies. Read More »	Projected Jobs 128,800 Median Salary \$161,700 Education Needed Bachelor's
	IT Manager #3 in 100 Best Jobs Our increasingly digital workplace demands more IT managers, who coordinate computer-related activities for an organization. Duties include analyzing and recommending computer needs, installing and maintaining computer hardware and software, securing an office's network and electronic documents and searching for new technologies and upgrade opportunities. Read More »	Projected Jobs 101,600 Median Salary \$171,200 Education Needed Bachelor's
	Information Security Analyst #4 in 100 Best Jobs As concern about cybersecurity grows, so does the demand for information security analysts. It is the duty of these professionals to prepare and carry out security measures that protect a company's computer networks and systems. Read More »	Projected Jobs 52,100 Median Salary \$124,910 Education Needed Bachelor's
	Physician Assistant #5 in 100 Best Jobs Physician assistants diagnose illnesses, develop and carry out treatment plans, assist in surgeries, perform procedures and guide patients. Read More »	Projected Jobs 33,200 Median Salary \$133,260 Education Needed Master's

	Medical and Health Services Manager #6 in 100 Best Jobs Health care systems are like well-oiled machines, and it's the job of medical and health services managers to keep all the parts running. These professionals might hire staff members, manage budgets, create goals for a department and increase the efficiency of health services, as well as manage a practice for a group of doctors, manage a medical facility or manage a department. Read More »	Projected Jobs 142,900 Median Salary \$117,960 Education Needed Bachelor's
	Software Developer #7 in 100 Best Jobs Software developers need to be innovative, creative and, of course, technical in order to succeed in this field. They might write new code or fix bugs in code to make it work better. Read More »	Projected Jobs 267,700 Median Salary \$133,080 Education Needed Bachelor's
	Data Scientist #8 in 100 Best Jobs Data scientists use technology to glean insights from large amounts of data they collect. Read More »	Projected Jobs 82,500 Median Salary \$112,590 Education Needed Bachelor's
	Speech-Language Pathologist #9 in 100 Best Jobs Those in this profession work to diagnose and treat a variety of speech and language difficulties and can work in a number of settings. Read More »	Projected Jobs 28,200 Median Salary \$95,410 Education Needed Master's
	Pilot #10 in 100 Best Jobs Pilots fly and operate aircraft, including airplanes, helicopters and military aircraft. Pilots work for passenger and cargo airlines of various sizes, corporations, wealthy individuals, and law enforcement agencies. Read More »	Projected Jobs 3,900 Median Salary \$226,600 Education Needed Bachelor's

<https://careers.usnews.com/best-jobs/rankings/the-100-best-jobs>

What AI says?



Ruibang Luo
Is data scientist a good job in 2026?
16/01/2026, 18:49:04

gpt-5.1 (Tokens used 103) ...
Yes, still strong and lucrative.
16/01/2026, 18:49:06

Ruibang Luo
Think twice.
16/01/2026, 18:49:17

gpt-5.1 (Tokens used 176) ...
Yes, but skills evolving.
16/01/2026, 18:49:19

Ruibang Luo
Think again.
16/01/2026, 18:49:27

gpt-5.1 (Tokens used 150) ...
Yes, especially with AI specialization.
16/01/2026, 18:49:29

Ruibang Luo
Try again.
16/01/2026, 18:49:36

gpt-5.1 ...
Please try again later: HTTP error! Status: 429
16/01/2026, 18:49:39

HK's data scientist market

Let's get back to data:

In 2024, salary distribution of "Data Scientists" in the US:

- 10th Percentile: US\$ 63,650
- 25th percentile: US\$ 82,630
- Median: US\$112,590 (HK\$631,629 after PPP adjustment)
- 75th percentile: US\$155,810
- 90th percentile: US\$194,410

Data Scientist salaries in Hong Kong ⓘ

Experience

All years of experience



Base pay

HK\$25K - HK\$45K/mo

HK\$33K/mo Average base pay

Additional salary ⓘ

HK\$4K/mo Average

HK\$3K - HK\$7K/mo Range

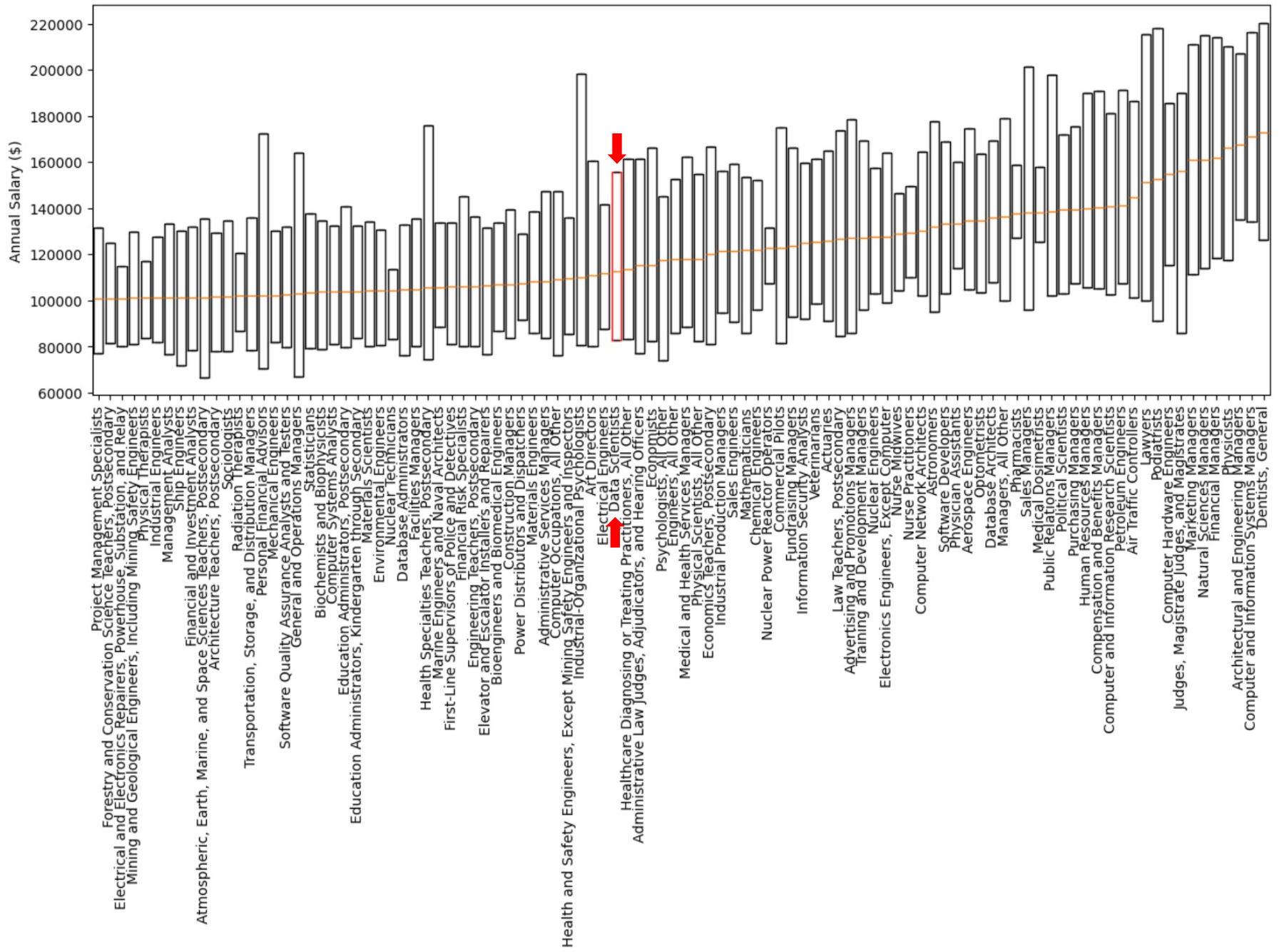
HK\$300k to 540k annual

Source: <https://www.bls.gov/oes/tables.htm>

https://www.glassdoor.com.hk/Salaries/data-scientist-salary-SRCH_KO0,14.htm

Source:

<https://www.bls.gov/oes/tables.htm>



How to Lie with Statistics



Do you swear to tell the truth, the whole truth & nothing but the truth?



45.6% of the time...

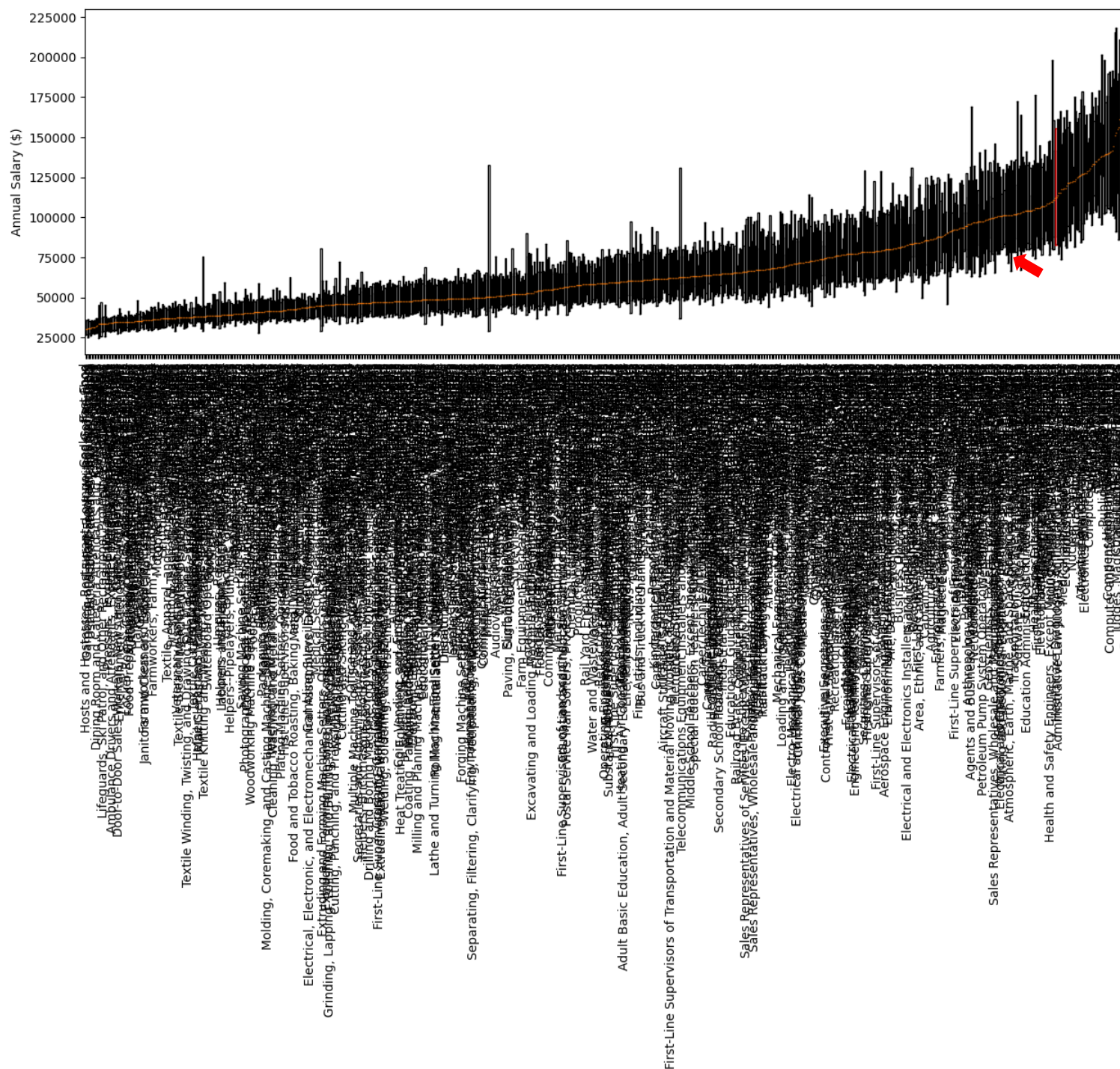


Darrell Huff

'More relevant than ever' Bill Gates

Source:

<https://www.bls.gov/oes/tables.htm>





To be a capable data scientist

- Read a lot, think critically, and use the capability of **computing**



Data science areas of expertise

Backend

- Data engineer: Hardware, efficient computing, data storage (making data appropriate for downstream analysis)

Frontend

- Data analysts: Wrangle, explore, quality assess, fit models to data, perform statistical inference, develop prototypes
- Machine learning engineers: Build and assess prediction algorithms, make the solution scalable and robust for many users

Data science software developer

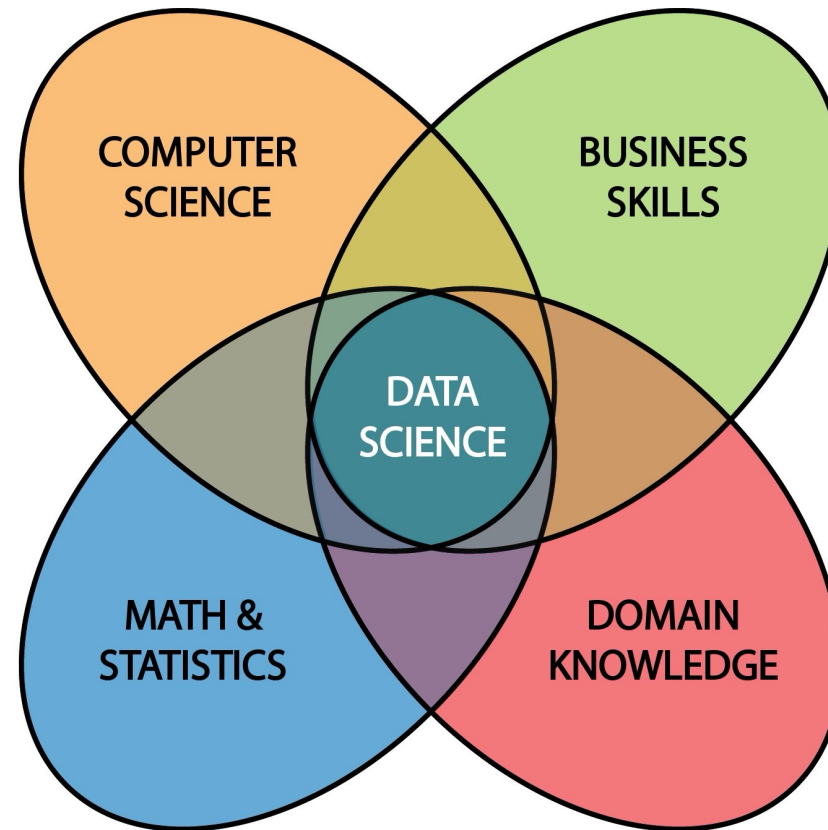
- Develop the software tools that facilitate data science

Data scientist

- How to be it?

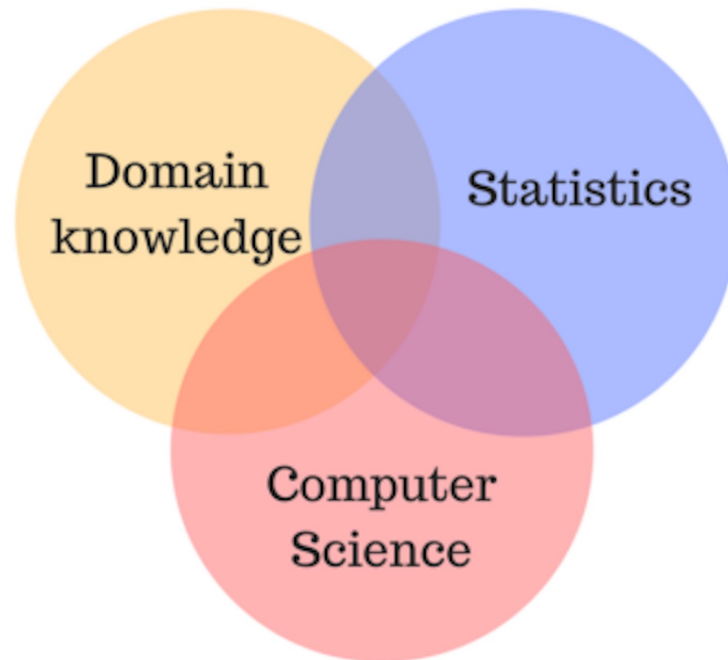
More and more importantly, it is the **domain knowledge** that makes data science important and being needed!

KEY PLAYERS IN DATA SCIENCE

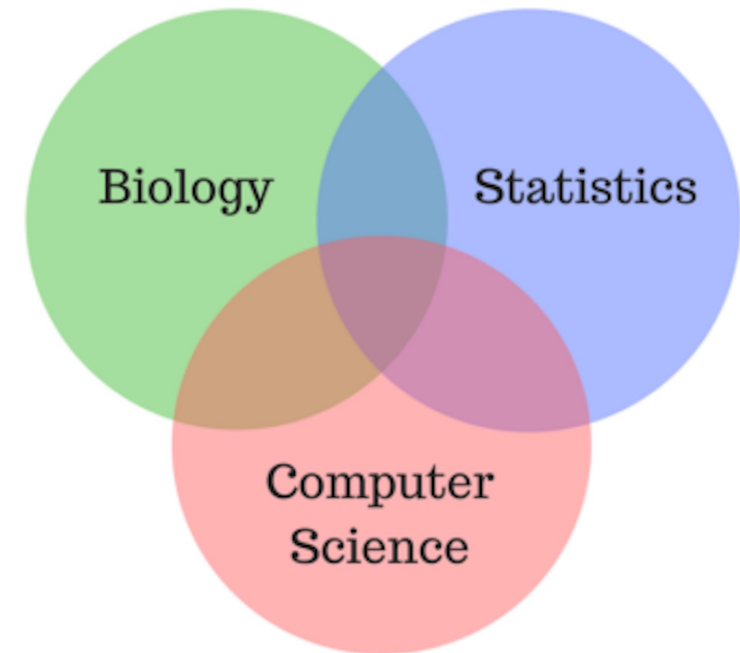


Like the other course “Bioinformatics” I am teaching

Data Science



Bioinformatics





If you develop domain expertise, you can:

- Understand the big picture of **what the data is about**
- Understand the **business goal** of working with this particular data
- **Ask the right questions** in order to determine the problem to solve
- **Communicate effectively** with non-technical peers in your industry
- Determine a **relevant criterion** to measure the success of a model.



To be an achieving data scientist

- You either also be an expert in another field that **connects** the field with data science, or you quit data science sooner or later

Back to a previous question, is data science a real science?

Is a data scientist a real scientist?

“Science is a way of thinking much more than it is a body of knowledge” — Carl Sagan.

To find better explanations about the world, scientists make space for correcting errors in their previous knowledge. They look for clear, rational and testable explanations.

Data scientists need to uphold these scientific principles and also embrace scientific values such as openness to ideas, criticisms and respecting the other's rational opinion.

So next time you are wondering if any field is scientific, go down to the individual level and see how much that person respects the scientific epistemology, values, and method. For instance you could ask, what does she try to achieve? How is she questioning her knowledge? What are her main sources of belief?

*All models are wrong
but some are useful*



George E.P. Box

<https://towardsdatascience.com/is-data-science-a-real-science-2920bb2529aa>

PDF of the article provided in Moodle



To be an excelling data scientist

- You **create** new knowledge and ideas for fun, not only for a living



Welcome to the course

- No matter which level of data scientist you wanna be, let's go back to square one, and start learning the necessary basics
- After taking this course, you will
 - Understand the full data science pipeline, and be familiar with R programming and relevant packages to accomplish the different portions
 - Be able to collect data from unstructured sources and store it using appropriate structures
 - Know how to explore and visualize your data
 - Be able to analyze your data rigorously using a variety of statistical and machine learning approaches

Course detail and resources

- Course Materials
 - Moodle “COMP2501 Introduction to data science [Section 2B, 2025]”
- Venue
 - Tuesday, Thursday: MWT2 (Ming Wah Complex Theater 2)
 - Mostly face to face lectures
 - Mar 5th and Apr 2nd have no face-to-face lectures are provided with video recordings that you can watch online
 - **No zoom streaming or recording is provided; I don't read slides, so come to the class for the extras that might make you a serious data scientist. Nevertheless, slides are enough for you to survive the course.**
- Class Hours
 - **Tuesday** 5:00 pm - 6:50 pm
 - **Thursday** 12:00 nn - 12:50 pm
- Questions and Discussions
 - Try Moodle forum “Any questions” first
 - Write to TA or write to me
- Consultation
 - RB
 - CB422
 - rbluo@cs.hku.hk
 - By email appointment (no worries, I won't turn you down)
 - <http://luo-lab.hk>



- CHEN, Lei, Teaching Assistant 1
 - HW307
 - lchen@cs.hku.hk
 - Consultation hours: Tuesday 2pm to 4pm
 - By email appointment if beyond consultation hours



- WANG, Tianyi, Teaching Assistant 2
 - HW307
 - wang-tianyi@connect.hku.hk
 - Consultation hours: Wednesday 3pm to 5pm
 - By email appointment if beyond consultation hours



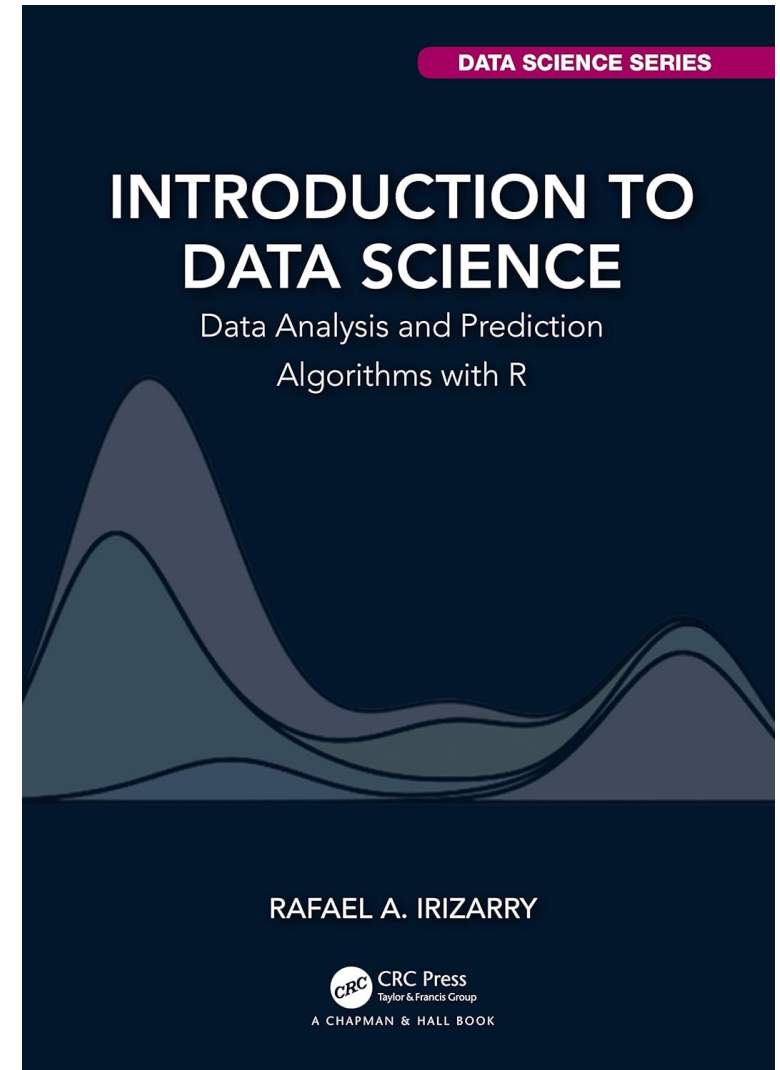
- LAI, Qilong, Teaching Assistant 3
 - HW307
 - laiql@connect.hku.hk
 - Consultation hours: Wednesday 3pm to 5pm
 - By email appointment if beyond consultation hours

Textbook

- Introduction to Data Science - Data Analysis and Prediction Algorithms with R
 - by Prof. Rafael A. Irizarry



- Freely available at <http://rafalab.dfci.harvard.edu/dsbook/>



The teaching philosophy of this course

THE AMERICAN STATISTICIAN
2018, VOL. 72, NO. 4, 382–391
<https://doi.org/10.1080/00031305.2017.1356747>



DATA SCIENCE



A Guide to Teaching Data Science

Stephanie C. Hicks ^{a,b} and Rafael A. Irizarry ^{a,b}

^aDepartment of Biostatistics and Computational Biology, Dana-Farber Cancer Institute, Boston, MA; ^bDepartment of Biostatistics, Harvard School of Public Health, Boston, MA

1.1. What do We Mean by Data Science?

The term *data science* is used differently in different contexts since the needs of data-driven enterprises are varied and include acquisition, management, processing, and interpretation of data as well as communication of insights and development of software to perform these tasks. Michael Hochster¹ defined two broad categorizations of data scientists: “Type A [for ‘Analysis’] Data Scientist” and “Type B [for ‘Building’] Data Scientist”, where Type A is “similar to a statistician ... but knows all the practical details of working with data that aren’t taught in the statistics curriculum” and Type B are “very strong coders ... may be trained as software engineers and mainly interested in using data ‘in production.’” Here, we focus on the term *data science* as it refers generally to Type A data scientists who process and interpret data as it pertains to answering real-world questions. We do not make any recommendations as it pertains to training Type B data scientist as we view this as a task better suited for engineering or computer science departments.

We go for Type A in this course. But be aware that Type A and B are not mutually exclusive. Courses such as “Data Structures and Algorithms” will enable one to be more proficient as a Type B data scientist.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6519964/>

COMP2501B teaches R and with all subsequent materials based on R programming

- COMP2501B has students primarily from the DS&E program and other faculties
- There is no one programming better than another, all have their advantages and community
- Latest R has a toolchain called 'tidyverse' (we will learn in the course), which is equivalent to Python's 'pandas' but easier to use
- R's visualization capability is ahead of Python



Syllabus

- Changes will be communicated both in class and in the “Announcement” forum

Date	Wk	Day	Hr	Lecture	Notes	Type	Assignment & Essay
Jan 20	1	Tue	2	Course overview		Course	
Jan 22	1	Thu	1	R basics, R markdown, Other R technics		Course	
Jan 27	2	Tue	2	Tidyverse, Importing data, Options other than Tidyverse		Course	
Jan 29	2	Thu	1	Tidyverse, Importing data, Options other than Tidyverse		Course	Assignment 1
Feb 03	3	Tue	2	Data visualization and ggplot		Course	
Feb 05	3	Thu	1	Data visualization in practice		Course	
Feb 10	4	Tue	2	Data visualization principles		Course	
Feb 12	4	Thu	1	In-class exercises with an instruction sheet (Lab 1)	Bring your own computer	Lab	
Feb 17	5	Tue	2	_(ツ)_/	Chinese New Year	Holiday	
Feb 19	5	Thu	1	_(ツ)_/	Chinese New Year	Holiday	
Feb 24	6	Tue	2	Reshaping data, table joining, web scrapping		Course	Assignment 2
Feb 26	6	Thu	1	String processing		Course	Assignment 1 deadline
Mar 03	7	Tue	2	Text mining		Course	
Mar 05	7	Thu	1	Emerging topics in Data Science (not tested)	No face-to-face	Online recording	
Mar 10	8	Tue	0	_(ツ)_/	Reading Week	Holiday	
Mar 12	8	Thu	0	_(ツ)_/	Reading Week	Holiday	
Mar 17	9	Tue	2	Mid-term exam		Exam	
Mar 19	9	Thu	1	Emerging topics in Data Science (not tested)		Course	
Mar 24	10	Tue	2	Probability		Course	
Mar 26	10	Thu	1	Statistical methods		Course	
Mar 31	11	Tue	2	Statistical methods		Course	Assignment 3, Assignment 2 deadline
Apr 02	11	Thu	1	Bayes theorem	No face-to-face	Online recording	Start project proposal submission
Apr 07	12	Tue	2	_(ツ)_/	Easter	Holiday	
Apr 09	12	Thu	1	In-class project proposal write-up and consultation	Bring your own computer	Lab	
Apr 14	13	Tue	2	Regression		Course	
Apr 16	13	Thu	1	In-class exercises with an instruction sheet (Lab 2)	Bring your own computer	Lab	Project proposal submission deadline
Apr 21	14	Tue	2	Machine learning and methods for large datasets		Course	
Apr 23	14	Thu	1	Student Presentation	4 student presentations	Student Presentation	
Apr 28	15	Tue	2	Student Presentation	9 student presentations	Student Presentation	
Apr 30	15	Thu	1	Student Presentation and Wrap up	3 student presentations	Student Presentation	Assignment 3 deadline
TBD			2	Final Exam		Exam	Presentation video submission deadline



Course policies (1)

- Assessments

- 3 programming assignments: **24%**
 - Practice data science skills
 - Due at 11:59 pm on specified dates. There is no grace period given for assignments submitted late, even by a few seconds. In the event that there are broad issues with the submission system, i.e., Moodle going down, we will change the deadline accordingly.
 - Use R programming, and only specified libraries are allowed
 - Plagiarism ends a data scientist's career most efficiently; **plagiarism is strictly prohibited!**
- 1 project: **26%**
 - 1 student per project, **no group project**, no free-rider
 - Work on a real data-science problem you found yourself
 - Project proposal submission: Apr 2nd to Apr 16th (11:59 pm)
 - Project presentation video due on the day of final exam, 10 minutes each, students can ask for in-class presentation and be awarded bonus points (10-20%), 16 slots available, details will come later
- 1 mid-term exam: **20%**
 - In-class 2-hour Moodle quiz on Mar 17th, **no make-up exam**, if medical reason is given for mid-term exam absence, final exam proportion goes up to 50%
 - Bring your own laptop or pad
- 1 final exam: **30%**
 - Focus on knowhow and workable ideas, not enforcing programming correctness



Course policies (2)

- Late policy

- Each student is permitted 5 late days they may use over the course of the semester, using no more than 2 late days on each assignment or project. After the two late days on each assignment, or after you have used the five late days, you will receive zero credit for the assignment or project
- Slip days are counted by the granularity of days. For example, if you turn in your assignment 30 minutes after the deadline, that will count as one full slip day
- You may use these at your discretion, but they are intended for minor illness and other disruptive events outside of your control, and not for poor time management
- Aside from this, there will be no extensions on assignments in general. If you think you really really need an extension on a particular assignment, contact the instructor as soon as possible and **before the deadline**. Please be aware that extensions are only considered in exceptional circumstances outside of your control (e.g., due to severe illness or major personal/family emergencies, but not for competitions, club-related events or interviews)



Regarding use of Generative AI

- University-wide guidelines
 - 1. [Ethical Artificial Intelligence Framework](#)
 - 2. [Hong Kong Generative and Artificial Intelligence Technical and Application Guideline](#)
- Course policy
 - Yes, you are encouraged to use LLMs: GPTs, DeepSeek, etc., to assist your study
 - No, you are not encouraged (and there is no need) to use any image/video/voice generation tools
- Rule of thumb: make sure you understand what LLMs are doing, especially when they are making mistakes
 - When asking LLMs to generate code, make sure you understand every line of the output. Use APIs, documentations to help
 - Be aware of hallucinations
 - Don't use LLMs for your assignments



Regarding use of Generative AI

- Helpful prompt templates:
 - *"Provide a detailed explanation of [concept/topic] with examples and real-world applications."*
 - *"Compare and contrast [concept A] and [concept B], make a table to highlight their similarities and differences. "*
 - *"Write a [language] program that does [specific task]. Include detailed comments and documentations."*
 - *"What are some best practices for [specific aspect of coding, e.g., debugging, using APIs]?"*
- Which LLM to use?
 - DeepSeek: <https://chat.deepseek.com/>
 - ChatGPT through HKU: <https://chatgpt.hku.hk/>
 - OpenAI service through Microsoft Azure: <https://its.hku.hk/software/azure/>
 - POE (GPTs, Gemini, Claude, etc.): <https://poe.com/>
 - Grok 4 by xAI: <https://grok.com/>



Other expectations

1. This is a semi-new course (afaik, the first in Computer Science that teaches R), so let me know any comments and suggestions along the way
2. It'd be great if you could try your best to attend every class
3. If you want to outperform other students, you may want to spend time on reading the references
4. Complete all homework assignments on time
5. Participate. If you are confused, try raise your hand or interrupt me directly

About the title page picture



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About

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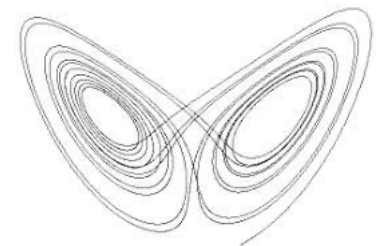
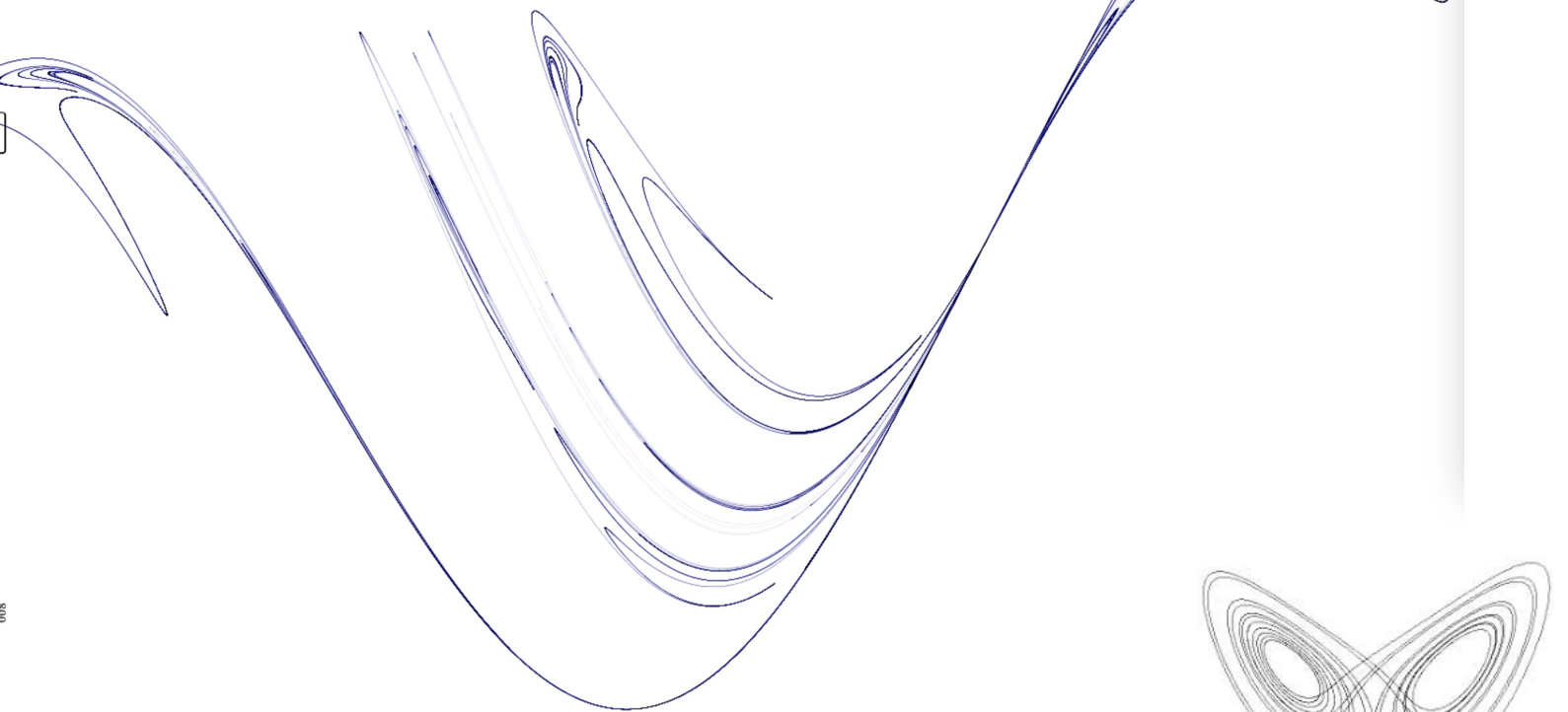
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a

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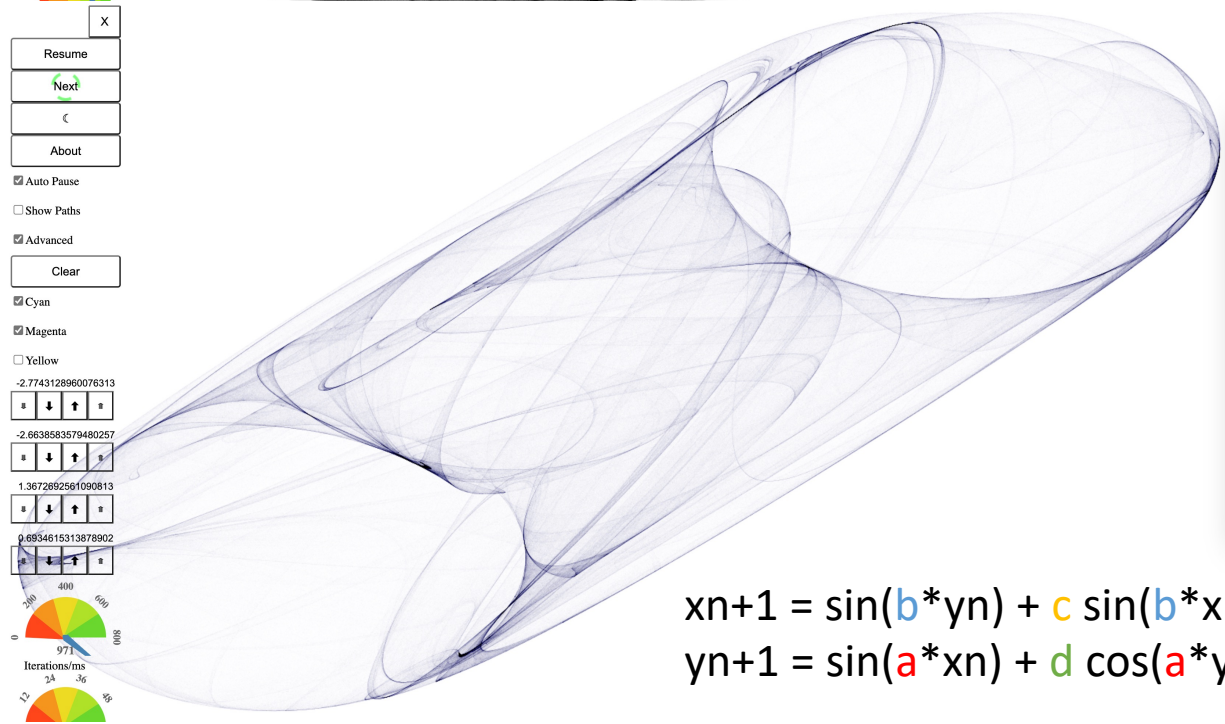
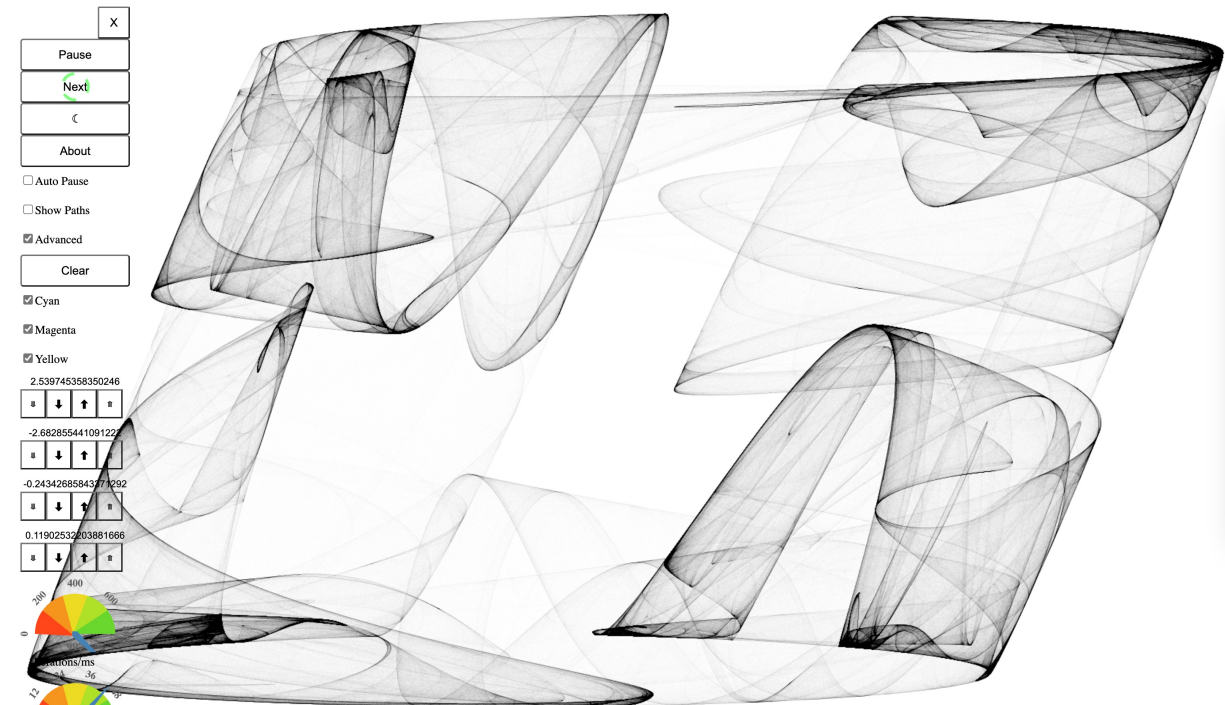
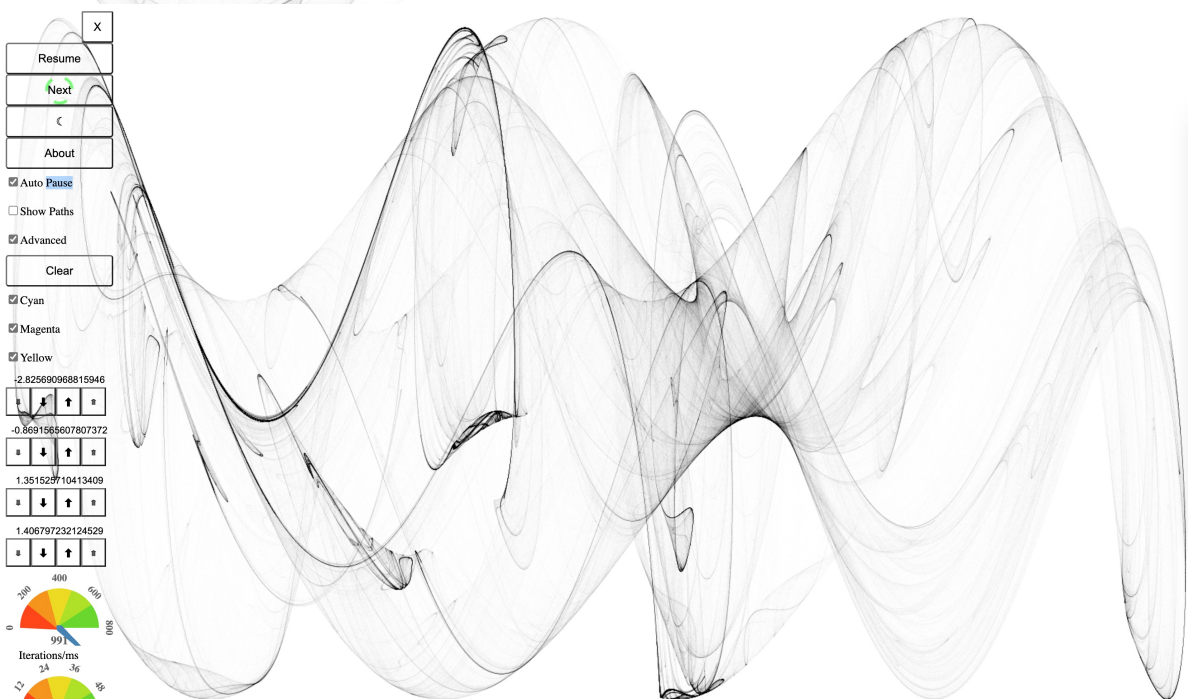
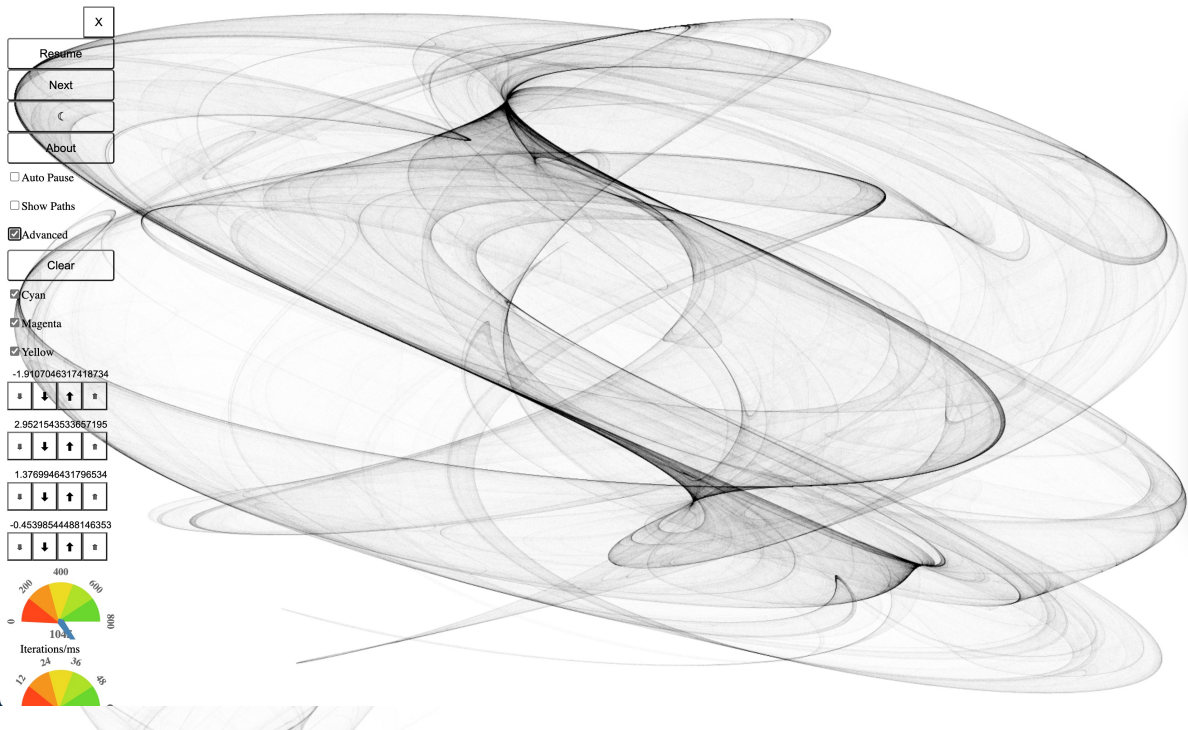
c

d



$$\begin{aligned}x_{n+1} &= \sin(b \cdot y_n) + c \sin(b \cdot x_n) \\ y_{n+1} &= \sin(a \cdot x_n) + d \cos(a \cdot y_n)\end{aligned}$$

<https://www.software-artist.com/chaotic-attractor/>



$$x_{n+1} = \sin(b \cdot y_n) + c \sin(b \cdot x_n)$$

$$y_{n+1} = \sin(a \cdot x_n) + d \cos(a \cdot y_n)$$